

SDITO - Lab

Name: Sanjib Ghara

Roll No.: CSE/23/065

Exam Roll No. 11600223092

Assignment 11

Problem Statement : Build scaling plans in AWS that balance the load on different EC2 instances.

Aim

To configure a Launch Template, create an Auto Scaling Group (ASG) integrated with an Application Load Balancer (ALB), and test both fault tolerance (instance termination) and scale-out policies (CPU stress).

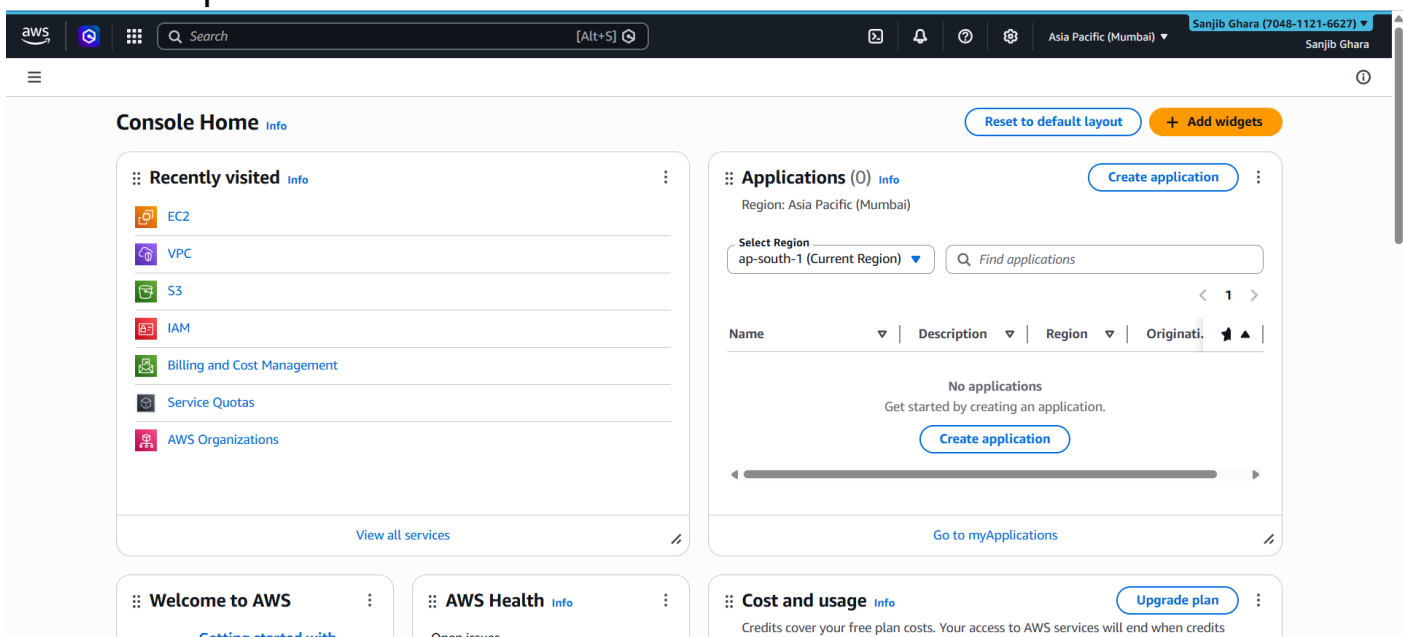
Architecture used

- EC2 Instances
- Target Group
- Application Load Balancer (ALB)
- Auto Scaling Group (ASG)
- Scaling Policy

Process:

Phase 1: Create a Launch Template

1. **Navigate to EC2 Dashboard:** Log in to the AWS Management Console and open the EC2 console.



2. **Access Launch Templates:** In the left navigation pane, under **Instances**, select **Launch Templates**, then click **Create launch template**.

EC2

Dashboard
AWS Global View
Events

▼ **Instances**
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager

▼ **Images**
AMIs
AMI Catalog

▼ **Elastic Block Store**
Volumes

Compute

EC2 launch templates

Streamline, simplify and standardize instance launches

Use launch templates to automate instance launches, simplify permission policies, and enforce best practices across your organization. Save launch parameters in a template that can be used for on-demand launches and with managed services, including EC2 Auto Scaling and EC2 Fleet. Easily update your launch parameters by creating a new launch template version.

New launch template

Create launch template

Benefits and features

Streamline provisioning
Minimize steps to provision instances. With EC2 Auto Scaling, updates to a launch template can be automatically

Simplify permissions
Create shorter, easier to manage IAM policies. [Learn more](#)

Documentation
[Documentation](#)
[API reference](#)

3. Template Details:

- **Name:** Enter **santemp065**.
- **Template version description:** **v0**
- Check the box for **Provide guidance to help set up a template that I can use with EC2 Auto Scaling**.

Create launch template

Creating a launch template allows you to create a saved instance configuration that can be reused, shared and launched at a later time. Templates can have multiple versions.

Launch template name and description

Launch template name - *required*

santemp065

Must be unique to this account. Max 128 chars. No spaces or special characters like '&', '"', '@'.

Template version description

v0

Max 255 chars

Auto Scaling guidance | [Info](#)
Select this if you intend to use this template with EC2 Auto Scaling

☒ Provide guidance to help me set up a template that I can use with EC2 Auto Scaling

► **Template tags**
► **Source template**

Summary

Software Image (AMI)
-

Virtual server type (instance type)
-

Firewall (security group)
-

Storage (volumes)
-

[Cancel](#) [Create launch template](#)

Launch template contents

Specify the details of your launch template below. Leaving a field blank will result in the field not being included in the launch template.

4. Application and OS Images (AMI): Search for and select **Ubuntu Server 24.04 LTS (HVM)**.

EC2 > **Launch templates** > Create launch template

Launch template contents

Specify the details of your launch template below. Leaving a field blank will result in the field not being included in the launch template.

▼ **Application and OS Images (Amazon Machine Image) - required** [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Recents **Quick Start**

Amazon Linux macOS Ubuntu Windows Red Hat SUSE Linux Debian

ami-05d2839d4773aafb (64-bit (x86)) / ami-0f6c99577b93a642b (64-bit (Arm))

Virtualization: hvm ENA enabled: true Root device type: ebs

Amazon Machine Image (AMI)

Ubuntu Server 24.04 LTS (HVM), SSD Volume Type

ami-05d2839d4773aafb (64-bit (x86)) / ami-0f6c99577b93a642b (64-bit (Arm))

Virtualization: hvm ENA enabled: true Root device type: ebs

Description

Ubuntu Server 24.04 LTS (HVM), EBS General Purpose (SSD) Volume Type. Support available from Canonical (<http://www.ubuntu.com/cloud/services>).

Summary

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd64...[read more](#)
ami-05d2839d4773aafb

Virtual server type (instance type)
-

Firewall (security group)
-

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Create launch template](#)

5. **Instance Type:** Select **t3.micro** (Free tier eligible).

6. **Key Pair:** Select an existing Key Pair (e.g., webkey1) or create a new RSA .pem key pair to ensure you can SSH into the instances later.

aws [Search] [Alt+S] Asia Pacific (Mumbai) Sanjib Ghara (7048-1121-6627) Sanjib Ghara

EC2 > Launch templates > Create launch template

▼ Instance type Info | Get advice Advanced

Instance type

t3.micro Free tier eligible

Family: t3 2 vCPU 1 GiB Memory Current generation: true
On-Demand Linux base pricing: 0.0112 USD per Hour On-Demand SUSE base pricing: 0.0112 USD per Hour
On-Demand Windows base pricing: 0.0204 USD per Hour On-Demand RHEL base pricing: 0.04 USD per Hour
On-Demand Ubuntu Pro base pricing: 0.0147 USD per Hour

Additional costs apply for AMIs with pre-installed software

☐ All generations Compare instance types

▼ Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name

webkey1 Create new key pair

▼ Network settings Info

Subnet Info

Don't include in launch template Create new subnet

When you specify a subnet, a network interface is automatically added to your template.

▼ Summary

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd64...read more
ami-05d2d839d4f73aafb

Virtual server type (instance type)
t3.micro

Firewall (security group)
-

Storage (volumes)
1 volume(s) - 8 GiB

Cancel Create launch template

7. Network Settings:

- **Subnet:** Leave this as "Don't include in launch template" (subnets will be defined in the ASG).
- **Firewall (Security Groups):** Select an existing security group (e.g., mykey1) that allows inbound traffic on port 22 (SSH) and port 4000 (Custom TCP).

aws [Search] [Alt+S] Asia Pacific (Mumbai) Sanjib Ghara (7048-1121-6627) Sanjib Ghara

EC2 > Launch templates > Create launch template

▼ Network settings Info

Subnet Info

Don't include in launch template Create new subnet

When you specify a subnet, a network interface is automatically added to your template.

Availability Zone Info

Don't include in launch template Enable additional zones

Not applicable for EC2 Auto Scaling

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Select existing security group ☐ Create security group

Security groups Info

Select security groups

mesec065 sg-0b5c6d3ad9d5946d9 VPC: vpc-02f5c07b934db25fd

► Advanced network configuration

▼ Storage (volumes) Info

EBS Volumes Hide details

▼ Summary

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd64...read more
ami-05d2d839d4f73aafb

Virtual server type (instance type)
t3.micro

Firewall (security group)
mesec065

Storage (volumes)
1 volume(s) - 8 GiB

Cancel Create launch template

8. **Advanced Details (User Data):** Scroll down to the User data field and paste the following bash script to automatically install necessary packages and start your Node.js application upon boot:

Bash

```
#!/bin/bash
apt-get update
apt-get upgrade -y
apt-get install -y nginx
systemctl start nginx
systemctl enable nginx
curl -sL https://deb.nodesource.com/setup_18.x | sudo -E bash -
apt-get install -y nodejs
git clone https://github.com/sanjib994/asg_9.git
cd awslab
npm install
node index.js
```

Allow tags in metadata [Info](#)

Don't include in launch template

User data - optional [Info](#)

Upload a file with your user data or enter it in the field.

[Choose file](#)

```
#!/bin/bash
apt-get update
apt-get upgrade -y
apt-get install -y nginx
systemctl start nginx
systemctl enable nginx
curl -sL https://deb.nodesource.com/setup_18.x | sudo -E bash -
apt-get install -y nodejs
git clone https://github.com/sanjib994/asg_9.git
cd asg_9
npm install
node index.js
```

☐ User data has already been base64 encoded

Summary

Software Image (AMI)
Canonical, Ubuntu, 24.04, amd64...[read more](#)
ami-05d2d839d4f73aafb

Virtual server type (instance type)
t3.micro

Firewall (security group)
mesec065

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Create launch template](#)

9. Create: Click **Create launch template**.

Success

Successfully created `santemp065(tt-0c4464cb04ee0b611)`.

► **Actions log**

Next Steps

Launch an instance

With On-Demand Instances, you pay for compute capacity by the second (for Linux, with a minimum of 60 seconds) or by the hour (for all other operating systems) with no long-term commitments or upfront payments. Launch an On-Demand Instance from your launch template.

[Launch instance from this template](#)

Create an Auto Scaling group from your template

Amazon EC2 Auto Scaling helps you maintain application availability and allows you to scale your Amazon EC2 capacity up or down automatically according to conditions you define. You can use Auto Scaling to help ensure that you are running your desired number of Amazon EC2 instances during demand spikes to maintain performance and decrease capacity during lulls to reduce costs.

[Create Auto Scaling group](#)

Create Spot Fleet

A Spot Instance is an unused EC2 instance that is available for less than the On-Demand price. Because Spot Instances enable you to request unused EC2 instances at steep discounts, you can lower your Amazon EC2 costs significantly. The hourly price for a Spot Instance (of each instance type in each Availability Zone) is set by Amazon EC2, and adjusted gradually based on the long-term supply of and demand for Spot Instances. Spot instances are well-suited for data-analysis, batch jobs, background processing, and optional tasks.

aws [Search] [Alt+S] Asia Pacific (Mumbai) Sanjib Ghara (7048-1121-6627) Sanjib Ghara

EC2 > Launch templates > santemp065

EC2

- Dashboard
- AWS Global View
- Events
- Instances
 - Instances
 - Instance Types
 - Launch Templates
 - Spot Requests
 - Savings Plans
 - Reserved Instances
 - Dedicated Hosts
 - Capacity Reservations
 - Capacity Manager
- Images
 - AMIs
 - AMI Catalog
- Elastic Block Store
 - Volumes

santemp065 (lt-0c4464cb04ee0b611)

Actions Delete template

Launch template details

Launch template ID lt-0c4464cb04ee0b611	Launch template name santemp065	Default version 1	Owner arn:aws:iam::704811216627:root
--	------------------------------------	----------------------	---

Details Versions Template tags

Launch template version details

Version: 1 (Default) Description: v0 Date created: 2026-04-04T13:00:49.000Z Created by: arn:aws:iam::704811216627:root

Actions Delete template version

AMI ID ami-05d2d839d4f73aafb	Instance type t3.micro	Availability Zone -	Availability Zone Id -
Key pair name webkey1	Security groups -	Security group IDs sg-0b5c6d3ad9d5946d9	

Phase 2: Create the Auto Scaling Group (ASG)

- 1. Navigate to Auto Scaling:** In the left panel, scroll down to **Auto Scaling** and select **Auto Scaling Groups**. Click **Create Auto Scaling group**.

aws [Search] [Alt+S] Asia Pacific (Mumbai) Sanjib Ghara (7048-1121-6627) Sanjib Ghara

EC2 > Auto Scaling groups

Amazon EC2 Auto Scaling

helps maintain the availability of your applications

Auto Scaling groups are collections of Amazon EC2 instances that enable automatic scaling and fleet management features. These features help you maintain the health and availability of your applications.

Create Auto Scaling group

Get started with EC2 Auto Scaling by creating an Auto Scaling group.

Create Auto Scaling group

How it works

Auto Scaling group

Pricing

Amazon EC2 Auto Scaling features have no additional fees beyond the service fees for Amazon EC2, CloudWatch (for scaling policies), and the other AWS resources that you use. Visit the pricing page of each service to learn more.

Getting started

- 2. Step 1: Choose Launch Template:** Name the AutoScalingGroup **sanas065**.
 - Select the **santemp065** your just created Launch template. Click **Next**.

aws [Search] [Alt+S] Asia Pacific (Mumbai) Sanjib Ghara (7048-1121-6627) Sanjib Ghara

EC2 > Auto Scaling groups > Create Auto Scaling group

Create Auto Scaling group

Step 1: Choose launch template

Step 2: Choose instance launch options

Step 3 - optional: Integrate with other services

Step 4 - optional: Configure group size and scaling

Step 5 - optional: Add notifications

Step 6 - optional: Add tags

Step 7: Review

Choose launch template

Specify a launch template that contains settings common to all EC2 instances that are launched by this Auto Scaling group.

Name: Auto Scaling group name
Enter a name to identify the group.
sanas065
Must be unique to this account in the current Region and no more than 255 characters.

Launch template info: For accounts created after May 31, 2023, the EC2 console only supports creating Auto Scaling groups with launch templates. Creating Auto Scaling groups with launch configurations is not recommended but still available via the CLI and API until December 31, 2023.

Launch template: Choose a launch template that contains the instance-level settings, such as the Amazon Machine Image (AMI), instance type, key pair, and security groups.
santemp065
Create a launch template

Version: Default (1)
Create a launch template version

EC2 > Auto Scaling groups > Create Auto Scaling group

santemp065

Create a launch template

Version: Default (1)

Create a launch template version

Description: v0

Launch template: santemp065 (lt-0c4464cb04ee0b611)

Instance type: t3.micro

AMI ID: ami-05d2d839d4f73aafb

Security groups: -

Request Spot Instances: No

Key pair name: webkey1

Security group IDs: sg-0b5c6d3ad9d5946d9

Additional details

Storage (volumes): -

Date created: Sat Apr 04 2026 18:30:49 GMT+0530 (India Standard Time)

Cancel Next

3. Step 2: Choose Instance Launch Options:

- Select your default **VPC**.
- Select multiple **Availability Zones** (e.g., ap-south-1a, ap-south-1b, ap-south-1c) for high availability. Click **Next**.

EC2 > Auto Scaling groups > Create Auto Scaling group

Step 1: Choose launch template

Step 2: Choose instance launch options

Step 3 - optional: Integrate with other services

Step 4 - optional: Configure group size and scaling

Step 5 - optional: Add notifications

Step 6 - optional: Add tags

Step 7: Review

Choose instance launch options

Choose the VPC network environment that your instances are launched into, and customize the instance types and purchase options.

Instance type requirements

You can keep the same instance attributes or instance type from your launch template, or you can choose to override the launch template by specifying different instance attributes or manually adding instance types.

Launch template: santemp065 (lt-0c4464cb04ee0b611)

Version: Default

Description: v0

Instance type: t3.micro

Override launch template

Network

For most applications, you can use multiple Availability Zones and let EC2 Auto Scaling balance your instances across the zones. The default VPC and default subnets are suitable for getting started quickly.

VPC

Choose the VPC that defines the virtual network for your Auto Scaling group.

vpc-02f5c07b934db25fd (172.31.0.0/16) Default

Create a VPC

EC2 > Auto Scaling groups > Create Auto Scaling group

vpc-02f5c07b934db25fd (172.31.0.0/16) Default

Create a VPC

Availability Zones and subnets

Define which Availability Zones and subnets your Auto Scaling group can use in the chosen VPC.

Select Availability Zones and subnets

aps1-az1 (ap-south-1a) | subnet-0b98e2f599c3265d0 (172.31.32.0/20) Default

aps1-az2 (ap-south-1c) | subnet-0e75f484b3572b0ff (172.31.16.0/20) Default

aps1-az3 (ap-south-1b) | subnet-06e624a16e70d45bf (172.31.0.0/20) Default

Create a subnet

Availability Zone distribution - new

Auto Scaling automatically balances instances across Availability Zones. If launch failures occur in a zone, select a strategy.

Balanced best effort

If launches fail in one Availability Zone, Auto Scaling will attempt to launch in another healthy Availability Zone.

Balanced only

If launches fail in one Availability Zone, Auto Scaling will continue to attempt to launch in the unhealthy Availability Zone to preserve balanced distribution.

Cancel Skip to review Previous Next

4. Step 3: Integrate with Other Services (Load Balancing):

- Select **Attach to a new load balancer**.
- Choose **Application Load Balancer** and set the scheme to **Internet-facing**.
- Under Listeners and routing, ensure the protocol is HTTP and set the Port to **4000**. Select **Create a target group** and name it → **sanas065-1**.
- Under Health Checks, set Health check grace period to 300 seconds . Click **Next**.

Load balancing [Info](#)

Use the options below to attach your Auto Scaling group to an existing load balancer, or to a new load balancer that you define.

Select Load balancing options

☐ No load balancer
Traffic to your Auto Scaling group will not be fronted by a load balancer.

☐ Attach to an existing load balancer
Choose from your existing load balancers.

☒ Attach to a new load balancer
Quickly create a basic load balancer to attach to your Auto Scaling group.

Attach to a new load balancer

Load balancer type
Choose from the load balancer types offered below. Type selection cannot be changed after the load balancer is created. If you need a different type of load balancer than those offered here, [visit the Load Balancing console](#).

☒ Application Load Balancer
HTTP, HTTPS

☐ Network Load Balancer
TCP, UDP, TLS

Load balancer name
Name cannot be changed after the load balancer is created.

sanas065-1

Must be unique within the Region. Use 1–32 alphanumeric characters and hyphens. Can't start with "internal-" or begin or end with a hyphen.

Load balancer scheme
Scheme cannot be changed after the load balancer is created.

☐ Internal

☒ Internet-facing

Network mapping
Your new load balancer will be created using the same VPC and Availability Zone selections as your Auto Scaling group. You can select different subnets and add subnets from additional Availability

VPC
[vpc-02f5c07b934db25fd](#)

Availability Zones and subnets
You must select a single subnet for each Availability Zone enabled. Only public subnets are available for selection to support DNS resolution.

☒ aps1-az1 (ap-south-1a) **Select a subnet**
subnet-0b98e2f599c3265d0

☒ aps1-az2 (ap-south-1c) **Select a subnet**
subnet-0e75f484b3572b0ff

☒ aps1-az3 (ap-south-1b) **Select a subnet**
subnet-06e624a16e70d45bf

Listeners and routing
If you require secure listeners, or multiple listeners, you can configure them from the [Load Balancing console](#) after your load balancer is created.

Protocol
HTTP

Port
4000

Default routing (forward to)
Create a target group

New target group name
An instance target group with default settings will be created.

sanas065-1

Must be unique within the Region. Use 1–32 alphanumeric characters and hyphens. Can't begin or end with a hyphen.

Tags - optional
Consider adding tags to your load balancer. Tags enable you to categorize your AWS resources so you can more easily manage them.

[Add tag](#)

aws [Search] [Alt+S] Asia Pacific (Mumbai) Sanjib Ghara (7048-1121-6627) Sanjib Ghara

EC2 > Auto Scaling groups > Create Auto Scaling group

☐ Enable zonal shift
New instance launches will be retargeted towards healthy Availability Zones until the zonal shift is canceled.

Health checks
Health checks increase availability by replacing unhealthy instances. When you use multiple health checks, all are evaluated, and if at least one fails, instance replacement occurs.

EC2 health checks
[Always enabled](#)

Additional health check types - optional [Info](#)

☐ Turn on Elastic Load Balancing health checks **Recommended**
Elastic Load Balancing monitors whether instances are available to handle requests. When it reports an unhealthy instance, EC2 Auto Scaling can replace it on its next periodic check.

☐ Turn on VPC Lattice health checks
VPC Lattice can monitor whether instances are available to handle requests. If it considers a target as failed a health check, EC2 Auto Scaling replaces it after its next periodic check.

☐ Turn on Amazon EBS health checks
EBS monitors whether an instance's root volume or attached volume stalls. When it reports an unhealthy volume, EC2 Auto Scaling can replace the instance on its next periodic health check.

Health check grace period [Info](#)
This time period delays the first health check until your instances finish initializing. It doesn't prevent an instance from terminating when placed into a non-running state.

300 seconds

Cancel Skip to review Previous **Next**

5. Step 4: Configure Group Size and Scaling:

- Set **Desired capacity** to 2, **Minimum capacity** to 2, and **Maximum capacity** to 3.
- Select **Target tracking scaling policy**.
- Metric type: **Average CPU utilization**.
- Target value: **20** (This means if average CPU usage goes above 20%, it will launch a new instance).
- Instance warmup: **300** seconds. Click **Next**.

aws [Search] [Alt+S] Asia Pacific (Mumbai) Sanjib Ghara (7048-1121-6627) Sanjib Ghara

EC2 > Auto Scaling groups > Create Auto Scaling group

Step 4 - optional
Configure group size and scaling

Step 5 - optional
Add notifications

Step 6 - optional
Add tags

Step 7
Review

Desired capacity type
Choose the unit of measurement for the desired capacity value. vCPUs and Memory(GiB) are only supported for mixed instances groups configured with a set of instance attributes.

Units (number of instances)

Desired capacity
Specify your group size.

2

Scaling [Info](#)
You can resize your Auto Scaling group manually or automatically to meet changes in demand.

Scaling limits
Set limits on how much your desired capacity can be increased or decreased.

Min desired capacity
2
Equal or less than desired capacity

Max desired capacity
3
Equal or greater than desired capacity

Automatic scaling - optional
Choose whether to use a target tracking policy [Info](#)
You can set up other metric-based scaling policies and scheduled scaling after creating your Auto Scaling group.

☒ No scaling policies
Your Auto Scaling group will remain at its initial size and will not dynamically resize to meet demand.

☐ Target tracking scaling policy
Choose a CloudWatch metric and target value and let the scaling policy adjust the desired capacity in proportion to the metric's value.

aws [Search] [Alt+S] Asia Pacific (Mumbai) Sanjib Ghara (7048-1121-6627) Sanjib Ghara

EC2 > Auto Scaling groups > Create Auto Scaling group

Equal or less than desired capacity Equal or greater than desired capacity

Automatic scaling - optional
Choose whether to use a target tracking policy [Info](#)
You can set up other metric-based scaling policies and scheduled scaling after creating your Auto Scaling group.

☐ No scaling policies
Your Auto Scaling group will remain at its initial size and will not dynamically resize to meet demand.

☒ Target tracking scaling policy
Choose a CloudWatch metric and target value and let the scaling policy adjust the desired capacity in proportion to the metric's value.

Scaling policy name
Target Tracking Policy

Metric type [Info](#)
Monitored metric that determines if resource utilization is too low or high. If using EC2 metrics, consider enabling detailed monitoring for better scaling performance.

Average CPU utilization

Target value
20

Instance warmup [Info](#)
300 seconds

☐ Disable scale in to create only a scale-out policy

aws

Search

[Alt+S]

Asia Pacific (Mumbai)

Sanjib Ghara (7048-1121-6627)

Sanjib Ghara

EC2

>

Auto Scaling groups

>

Create Auto Scaling group

Additional settings

Instance scale-in protection

If protect from scale in is enabled, newly launched instances will be protected from scale in by default.

☐ Enable instance scale-in protection

Monitoring

Info

☐ Enable group metrics collection within CloudWatch

Default instance warmup

Info

The amount of time that CloudWatch metrics for new instances do not contribute to the group's aggregated instance metrics, as their usage data is not reliable yet.

☐ Enable default instance warmup

Auto Scaling group deletion protection - new

Info

Configure how this Auto Scaling group can be deleted.

☒ None (default)

The Auto Scaling group can be deleted.

☐ Prevent force deletion

The Auto Scaling group cannot be force deleted. To delete the group, all instances must be detached or terminated, and all warm pools must be deleted.

☐ Prevent all deletion

The Auto Scaling group cannot be deleted.

Cancel

Skip to review

Previous

Next

6. **Finalize:** Skip Step 5 (Notifications) and Step 6 (Tags) by clicking Next. Review your configurations and click **Create Auto Scaling group**.

aws

Search

[Alt+S]

Asia Pacific (Mumbai)

Sanjib Ghara (7048-1121-6627)

Sanjib Ghara

EC2

>

Auto Scaling groups

>

Create Auto Scaling group

Step 1
Choose launch template

Step 2
Choose instance launch options

Step 3 - optional
Integrate with other services

Step 4 - optional
Configure group size and scaling

Step 5 - optional
Add notifications

Step 6 - optional
Add tags

Step 7
Review

Add notifications - optional

Info

Send notifications to SNS topics whenever Amazon EC2 Auto Scaling launches or terminates the EC2 instances in your Auto Scaling group.

Add notification

Cancel

Skip to review

Previous

Next

aws

Search

[Alt+S]

Asia Pacific (Mumbai)

Sanjib Ghara (7048-1121-6627)

Sanjib Ghara

EC2

>

Auto Scaling groups

>

Create Auto Scaling group

Step 1
Choose launch template

Step 2
Choose instance launch options

Step 3 - optional
Integrate with other services

Step 4 - optional
Configure group size and scaling

Step 5 - optional
Add notifications

Step 6 - optional
Add tags

Step 7
Review

Add tags - optional

Info

Add tags to help you search, filter, and track your Auto Scaling group across AWS. You can also choose to automatically add these tags to instances when they are launched.

You can optionally choose to add tags to instances (and their attached EBS volumes) by specifying tags in your launch template. We recommend caution, however, because the tag values for instances from your launch template will be overridden if there are any duplicate keys specified for the Auto Scaling group.

Tags (0)

No tags

Add tag

50 remaining

Cancel

Previous

Next

Capacity Reservation preference

Preference Default	Capacity Reservation IDs -	Resource Groups -
-----------------------	-------------------------------	----------------------

Step 5: Add notifications Edit

Notifications
No notifications

Step 6: Add tags Edit

Tags (0)

Key	Value	Tag new instances
No tags		

[Preview code](#) [Cancel](#) [Previous](#) [Create Auto Scaling group](#)

After waiting some time....

Instances (3) Info Connect Instance state Actions Launch instances

Saved filter sets
Choose filter set Find Instance by attribute or tag (case-sensitive)

☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4
☐		i-0239a4c141696b1ef	Running	t3.micro	3/3 checks passed	View alarms	ap-south-1b	ec2-3-110-1
☐		i-0d042ac49c90394cb	Running	t3.micro	3/3 checks passed	View alarms	ap-south-1c	ec2-13-203
☐	sanjib_asg12	i-0d4d65be8ca2f53d7	Terminated	t3.micro	-	View alarms	ap-south-1a	-

Select an instance

Phase 3: Verify Application Deployment

Ensure the ASG successfully launched instances and the Node.js app is running.

1. Go back to the **Instances** dashboard. I should see two instances in the "Initializing" or "Running" state.
2. Select the first instance, copy its **Public IPv4 address**, and paste it into a web browser followed by port 4000 (e.g., `http://<Public-IP>:4000`). It should display **"Hello MCKV"**.
3. Repeat the process for the second instance to verify it is also functioning correctly.

Open 1st instance and Copy Public IPv4 address and Paste it in browser:

Instance summary for i-0239a4c141696b1ef Info

Updated less than a minute ago

Instance ID i-0239a4c141696b1ef	Public IPv4 address 3.110.155.47 open address	Private IPv4 addresses 172.31.2.5
IPv6 address -	Instance state Running	Public DNS ec2-3-110-155-47.ap-south-1.compute.amazonaws.com open address
Hostname type IP name: ip-172-31-2-5.ap-south-1.compute.internal	Private IP DNS name (IPv4 only) ip-172-31-2-5.ap-south-1.compute.internal	Elastic IP addresses -
Answer private resource DNS name -	Instance type t3.micro	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendation s. Learn more
Auto-assigned IP address 3.110.155.47 [Public IP]	VPC ID vpc-02f5c07b934db25fd	Auto Scaling Group name sanas065
IAM role -	Subnet ID subnet-06e624a16e70d45bf	Managed false
IMDSv2 Required	Instance ARN arn:aws:ec2:ap-south-1:704811216627:instance/i-0239a4c141696b1ef	

sanjib994/asg_9 x Instance details | EC2 | ap-south-1 x 3.110.155.47:4000 x +

Not secure 3.110.155.47:4000

Hello MCKV

Open 2nd instance and Copy Public IPv4 address and Paste it in browser:

Instance summary for i-0d042ac49c90394cb Info

Updated less than a minute ago

Instance ID i-0d042ac49c90394cb	Public IPv4 address 13.203.47.159 open address	Private IPv4 addresses 172.31.25.130
IPv6 address -	Instance state Running	Public DNS ec2-13-203-47-159.ap-south-1.compute.amazonaws.com open address
Hostname type IP name: ip-172-31-25-130.ap-south-1.compute.internal	Private IP DNS name (IPv4 only) ip-172-31-25-130.ap-south-1.compute.internal	Elastic IP addresses -
Answer private resource DNS name -	Instance type t3.micro	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendation s. Learn more
Auto-assigned IP address 13.203.47.159 [Public IP]	VPC ID vpc-02f5c07b934db25fd	Auto Scaling Group name sanas065
IAM role -	Subnet ID subnet-0e75f484b3572b0ff	Managed false
IMDSv2 Required	Instance ARN arn:aws:ec2:ap-south-1:704811216627:instance/i-0d042ac49c90394cb	

sanjib994/asg_9 x Instance details | EC2 | ap-south-1 x 3.110.155.47:4000 x 13.203.47.159:4000 x +

Not secure 13.203.47.159:4000

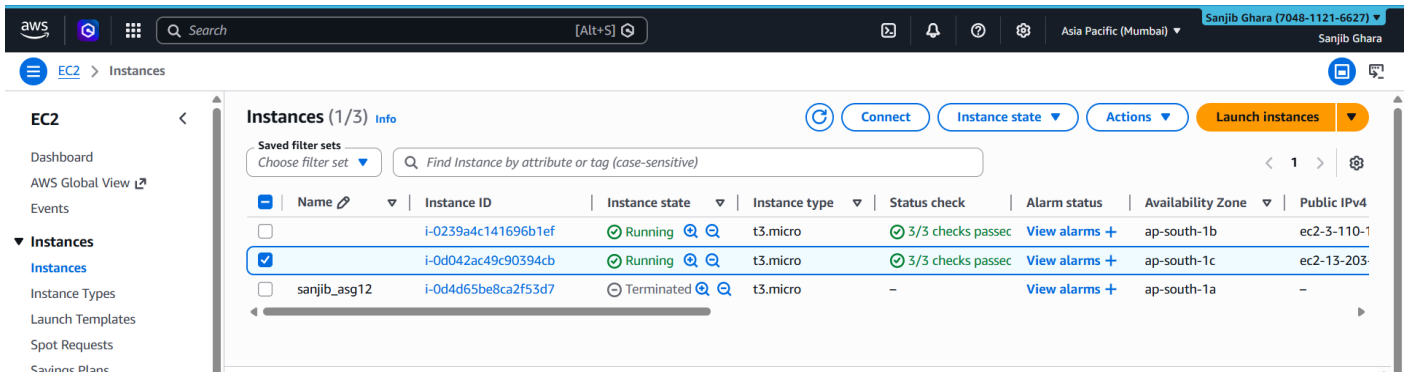
Hello MCKV

Phase 4: Test Auto-Recovery (Fault Tolerance)

Test if the ASG replaces crashed or terminated instances.

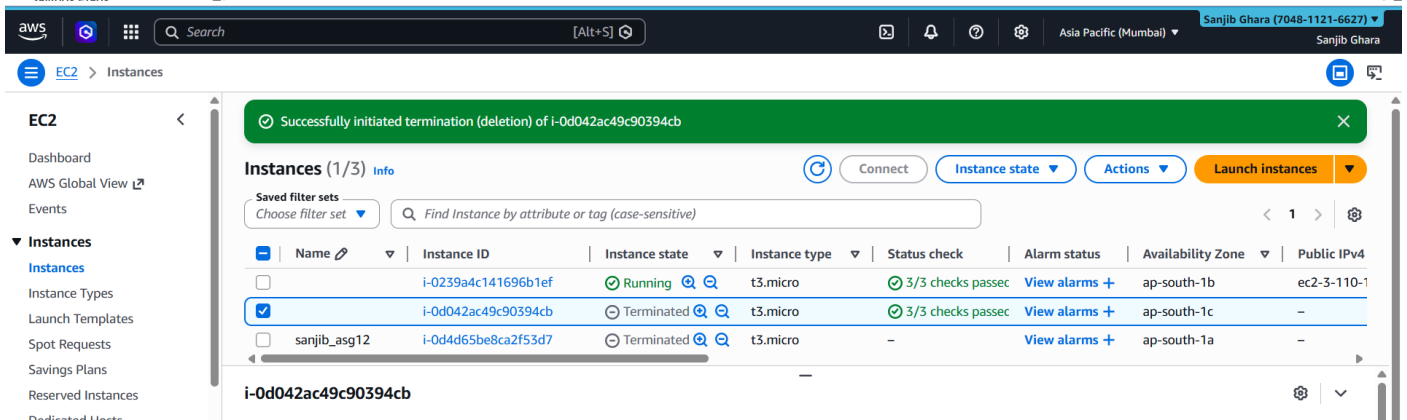
1. In the EC2 Instances dashboard, select one of the two running instances.
2. Go to **Instance state** → **Terminate (delete) instance** and confirm.
3. **Observation:** Wait a few moments. The Auto Scaling Group will detect that the instance count has fallen below the minimum desired capacity (2) and will

4. automatically initiate a new server to replace the terminated one.



Instances (1/3) Info

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4
	i-0239a4c141696b1ef	Running	t3.micro	3/3 checks passed	View alarms +	ap-south-1b	ec2-3-110-1
☒	i-0d042ac49c90394cb	Running	t3.micro	3/3 checks passed	View alarms +	ap-south-1c	ec2-13-203
	sanjib_asg12	Terminated	t3.micro	-	View alarms +	ap-south-1a	-



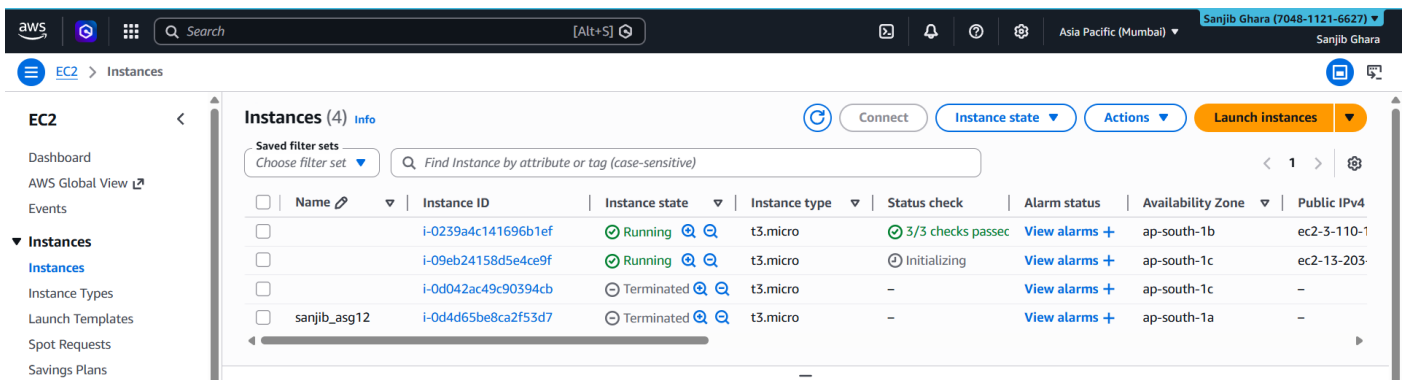
Successfully initiated termination (deletion) of i-0d042ac49c90394cb

Instances (1/3) Info

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4
	i-0239a4c141696b1ef	Running	t3.micro	3/3 checks passed	View alarms +	ap-south-1b	ec2-3-110-1
☒	i-0d042ac49c90394cb	Terminated	t3.micro	3/3 checks passed	View alarms +	ap-south-1c	-
	sanjib_asg12	Terminated	t3.micro	-	View alarms +	ap-south-1a	-

i-0d042ac49c90394cb

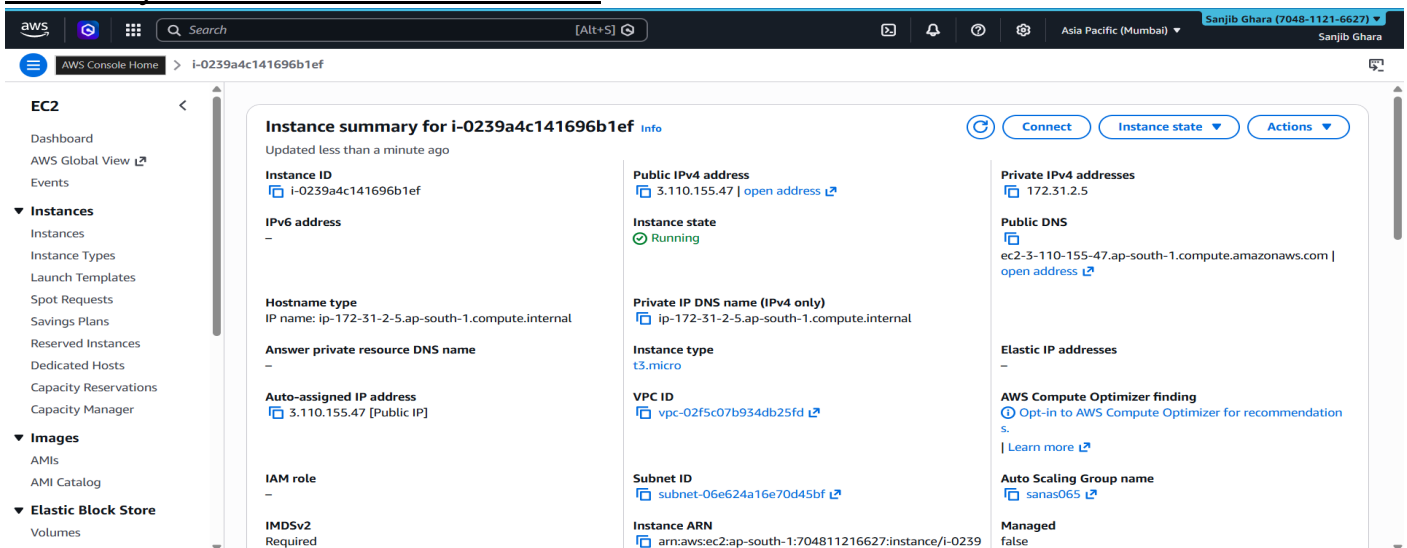
The Auto Scaling Group automatically provisions a replacement server after an instance fails or becomes overloaded.



Instances (4) Info

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public IPv4
	i-0239a4c141696b1ef	Running	t3.micro	3/3 checks passed	View alarms +	ap-south-1b	ec2-3-110-1
	i-09eb24158d5e4ce9f	Running	t3.micro	Initializing	View alarms +	ap-south-1c	ec2-13-203
☒	i-0d042ac49c90394cb	Running	t3.micro	-	View alarms +	ap-south-1c	-
	sanjib_asg12	Terminated	t3.micro	-	View alarms +	ap-south-1a	-

Click any one of the two instances :



Instance summary for i-0239a4c141696b1ef Info

Updated less than a minute ago

Instance ID: i-0239a4c141696b1ef

Public IPv4 address: 3.110.155.47 | open address

Instance state: Running

Private IP DNS name (IPv4 only): ip-172-31-2-5.ap-south-1.compute.internal

Instance type: t3.micro

VPC ID: vpc-02f5c07b934db25fd

Subnet ID: subnet-06e624a16e70d45bf

Instance ARN: arn:aws:ec2:ap-south-1:704811216627:instance/i-0239

Private IPv4 addresses: 172.31.2.5

Public DNS: ec2-3-110-155-47.ap-south-1.compute.amazonaws.com | open address

Elastic IP addresses: -

AWS Compute Optimizer finding: Opt-in to AWS Compute Optimizer for recommendation s. | Learn more

Auto Scaling Group name: sanas065

Managed: false

Phase 5: Test Scale-Out Policy (CPU Stress Test)

Artificially spike the CPU to trigger the target tracking scaling policy.

1. Select one of the running instances and connect to it using an SSH client (e.g., Bitwise SSH Client) using your .pem key.
2. Accept the Host Key Verification prompt.
3. Open a new terminal console within Bitwise and create a script to stress the CPU:
 - Type `nano file1.sh`.
 - Write the following infinite loop script:

Bash

```
#!/bin/bash
```

```
while true
```

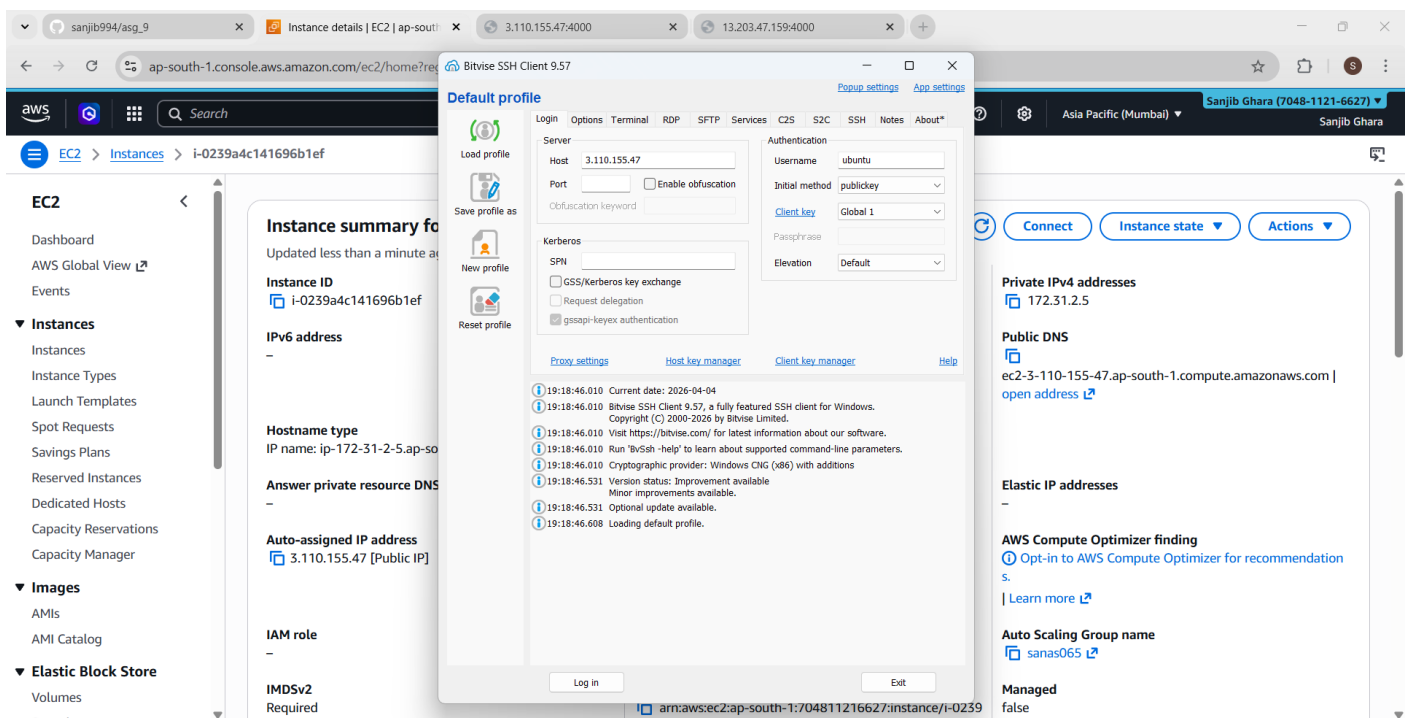
```
do
```

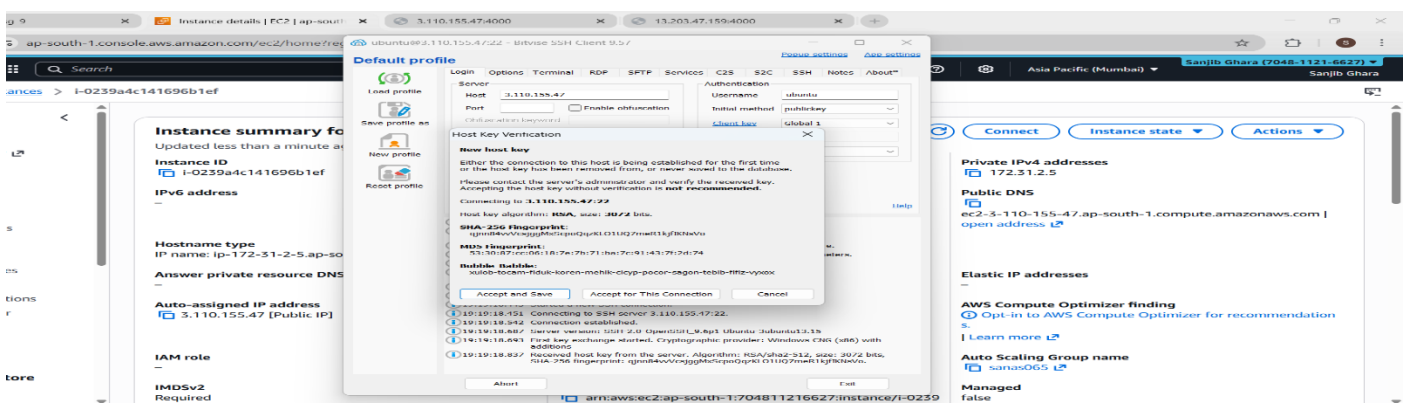
```
echo "Loop Running..."
```

```
done
```

- Save and exit (Ctrl+O, Enter, Ctrl+X).

4. Make the script executable by running `chmod +x file1.sh`.
5. Execute the script by running `./file1.sh`.
6. **Observation:** Go back to the AWS Console and check the EC2 metrics. You will notice the CPU utilization spiking significantly. Because it crosses your target value of 20%, the ASG will trigger an alarm and automatically launch a 3rd instance to handle the load.
7. Log out of the Bitwise SSH session to stop the script.

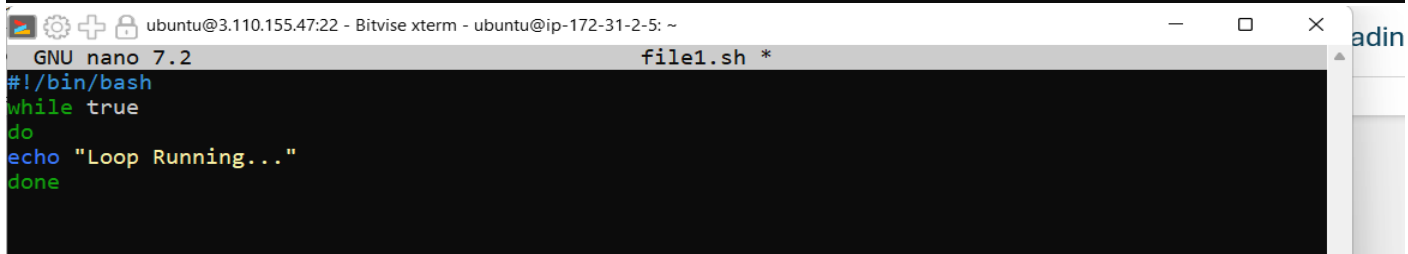




Open New Terminal Console:

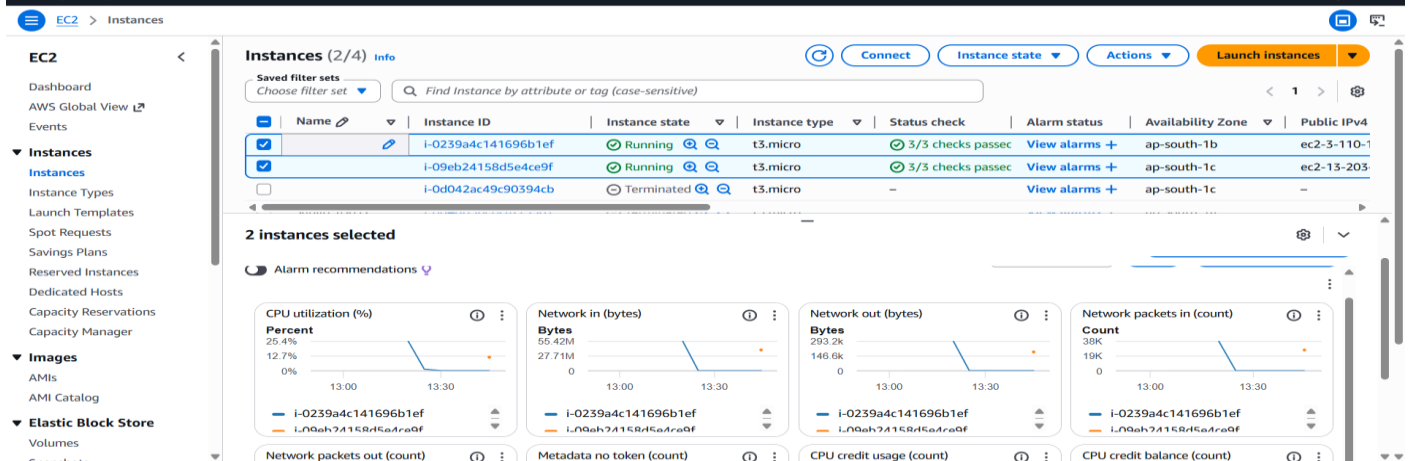
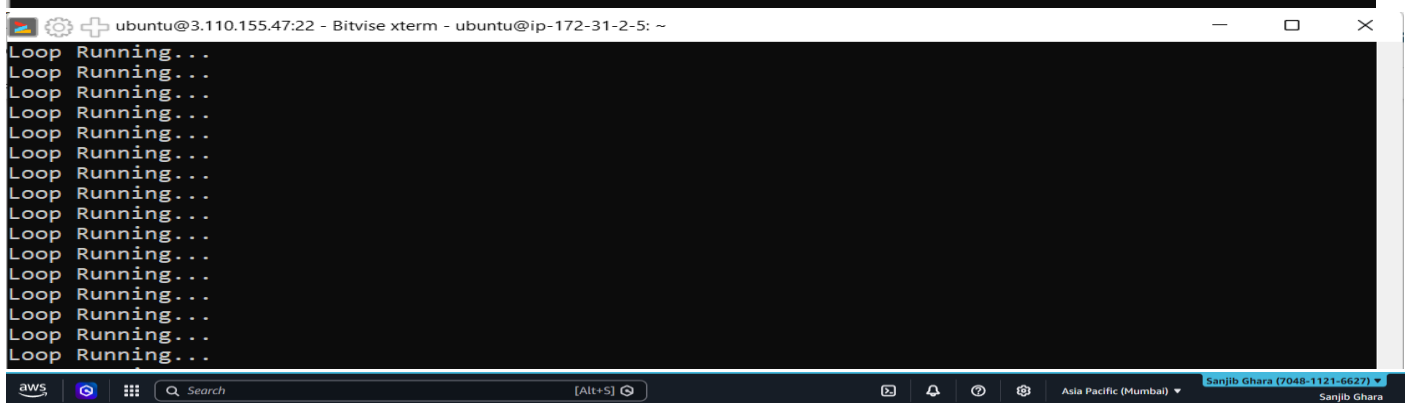
```
Last login: Sat Apr 4 13:49:43 2026 from 152.58.183.119
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

ubuntu@ip-172-31-2-5:~$ nano file1.sh
```

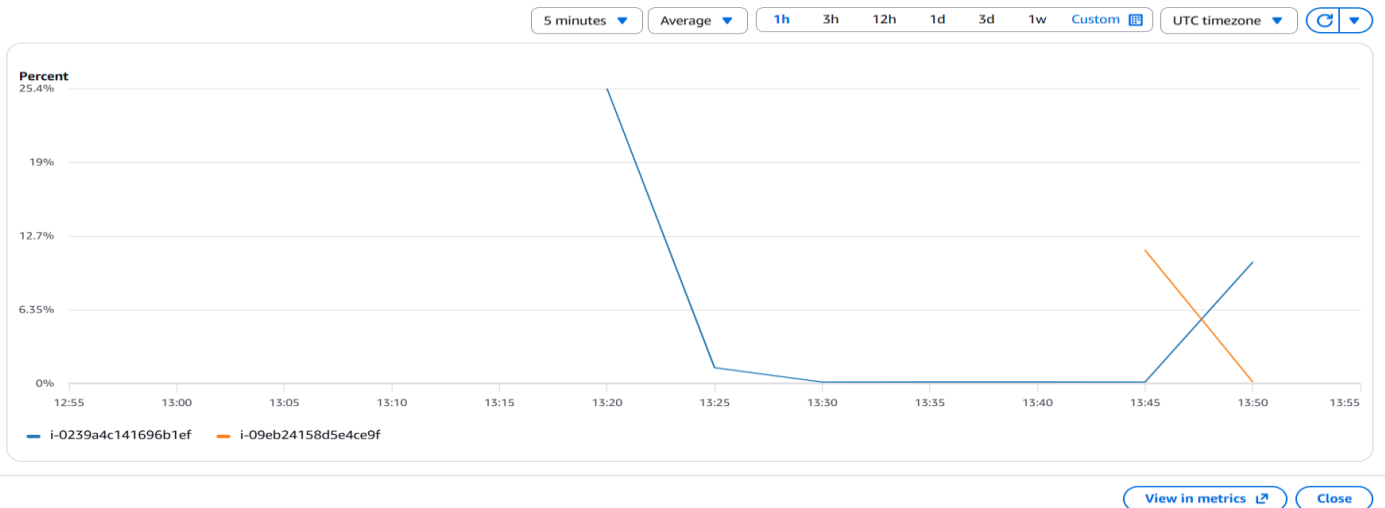


Press Ctrl +x to Exit...

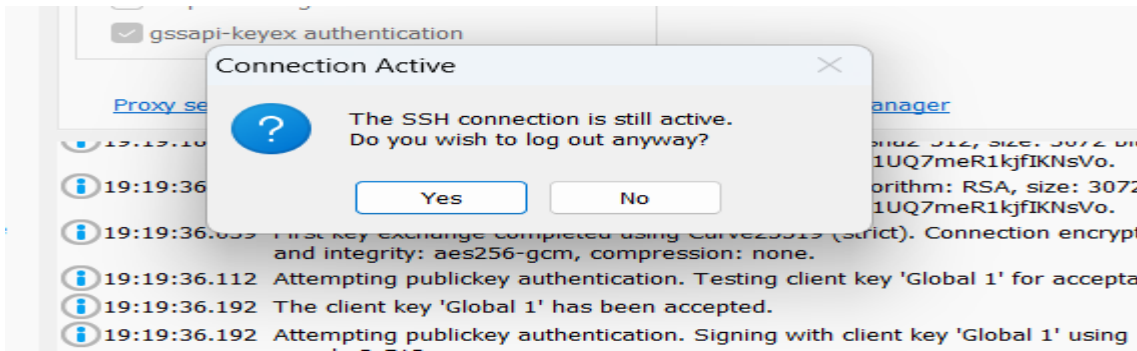
```
ubuntu@ip-172-31-2-5:~$ nano file1.sh
ubuntu@ip-172-31-2-5:~$ chmod +x file1.sh
ubuntu@ip-172-31-2-5:~$ ./file1.sh
```



CPU utilization (%)



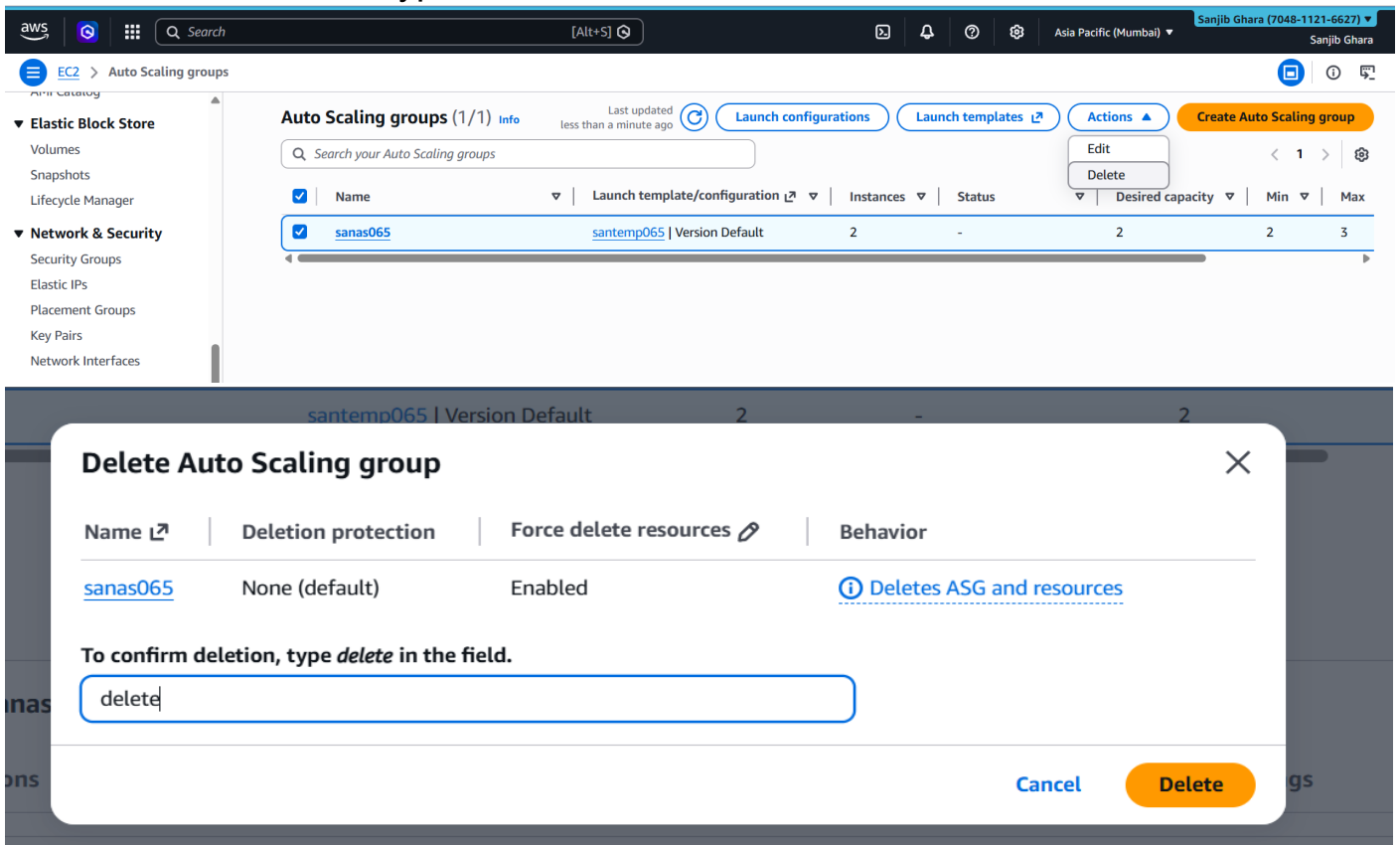
Now Log out from the bitwise SSH ..



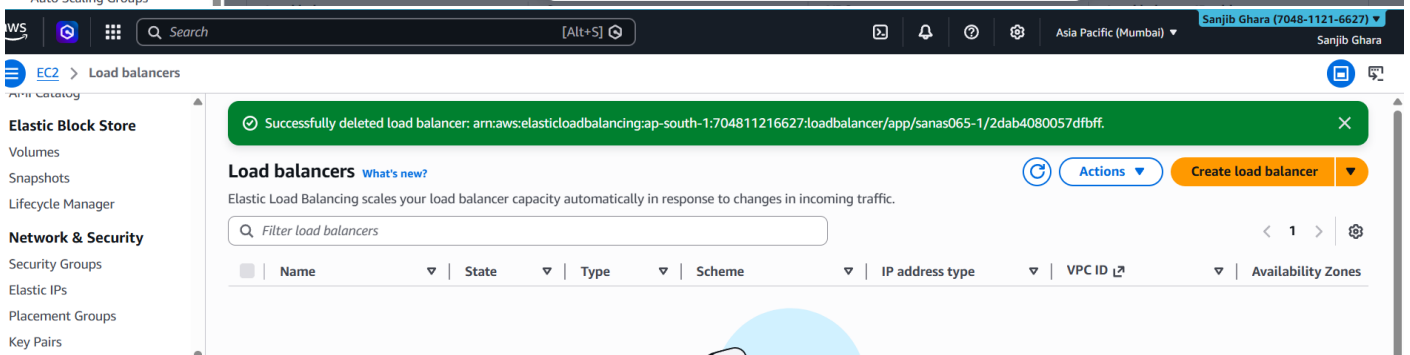
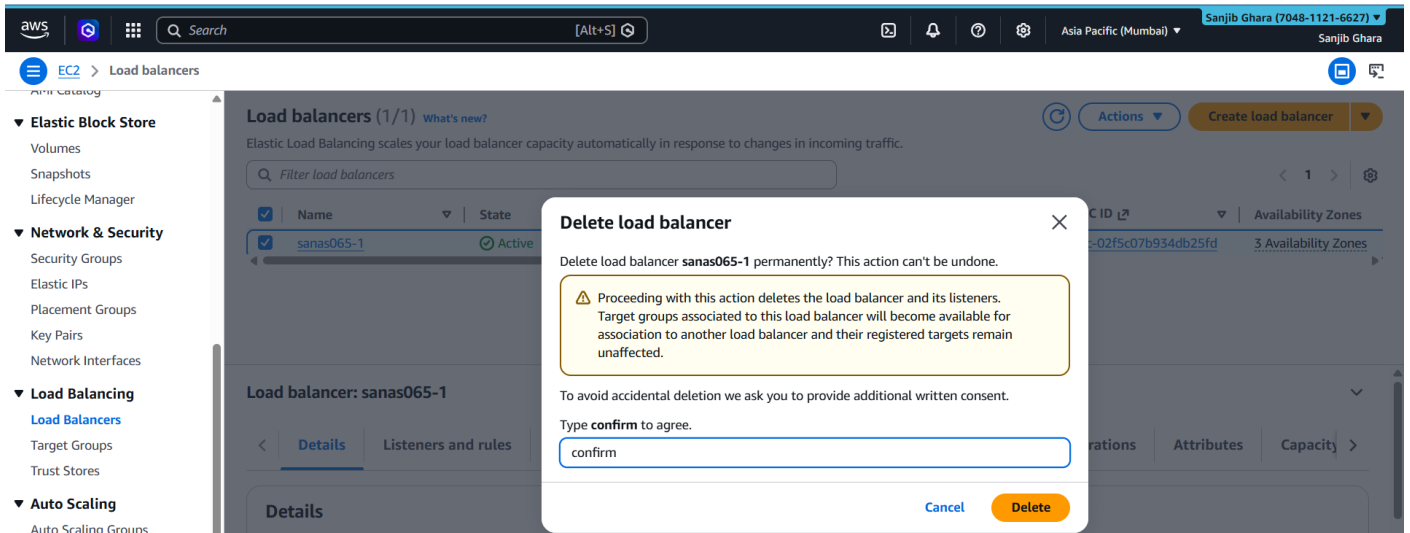
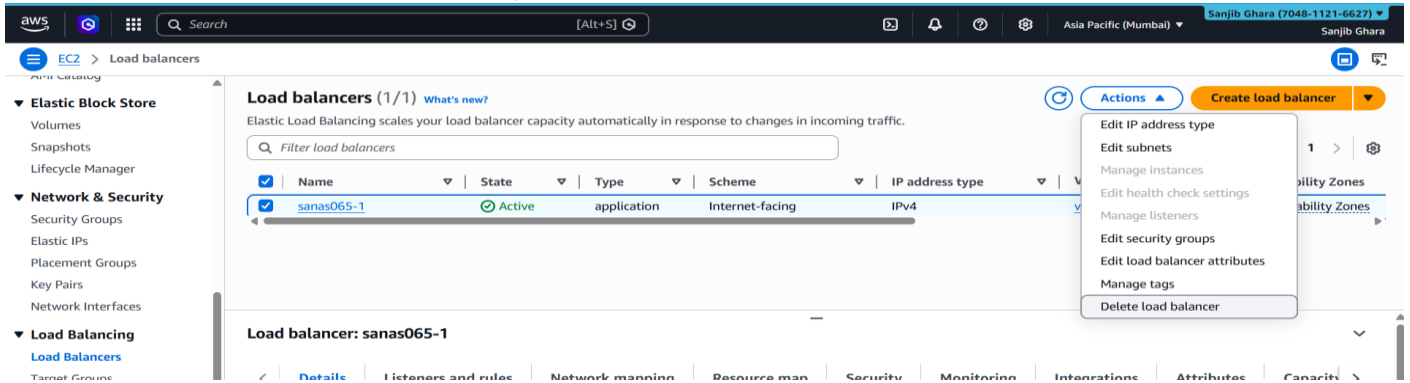
Phase 6: Resource Cleanup

Crucial step to ensure you do not incur unwanted AWS charges.

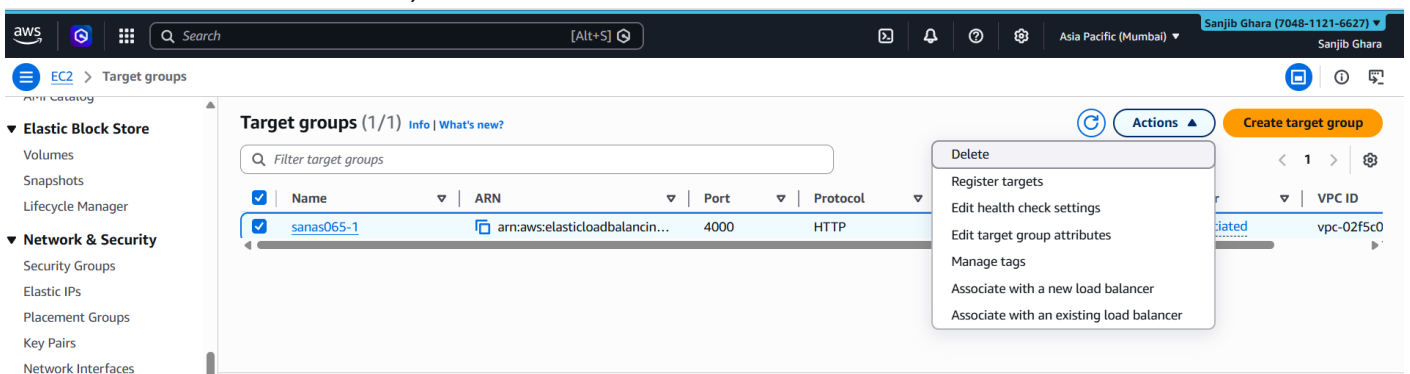
1. **Delete Auto Scaling Group:** Go to **Auto Scaling Groups**, select **sanas065**, click **Delete**, and type "delete" to confirm.

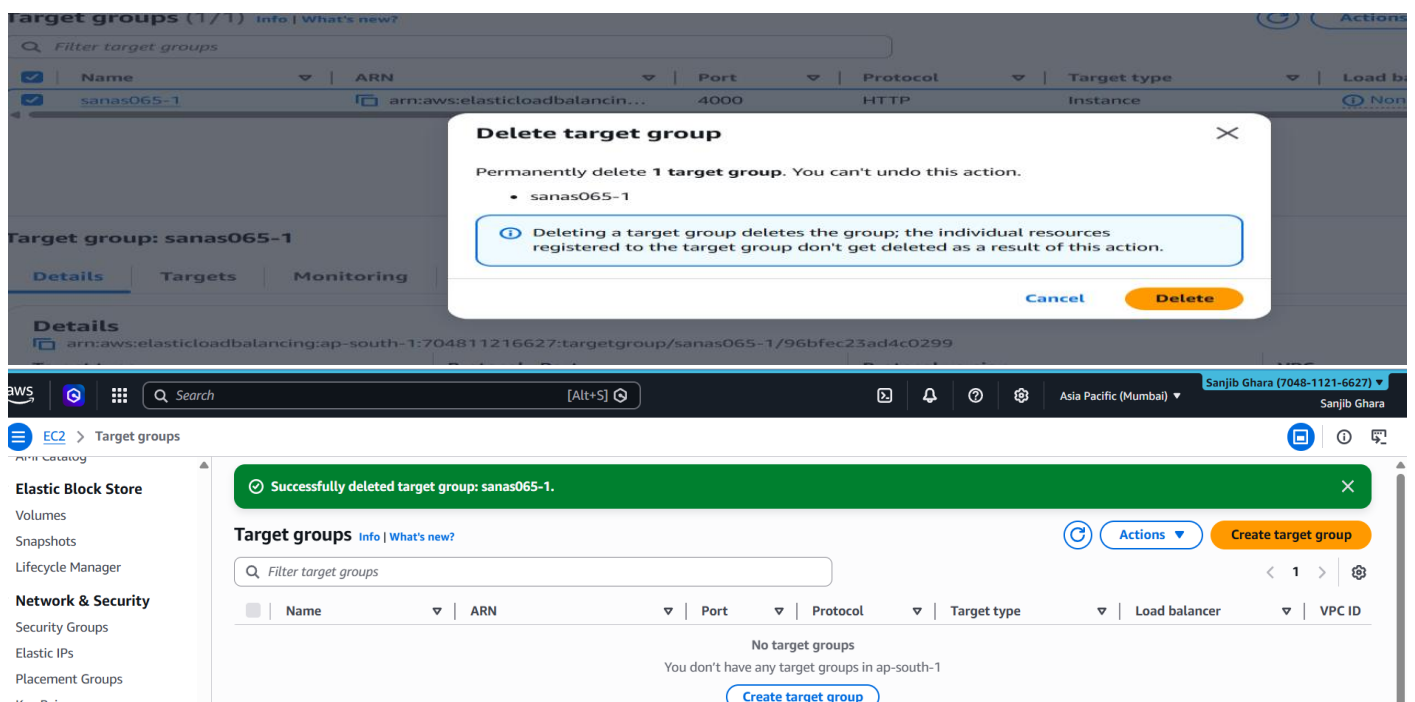


2. Delete Load Balancer: Go to Load Balancers, select **sanas065-1**, click **Actions → Delete**, and type "confirm".

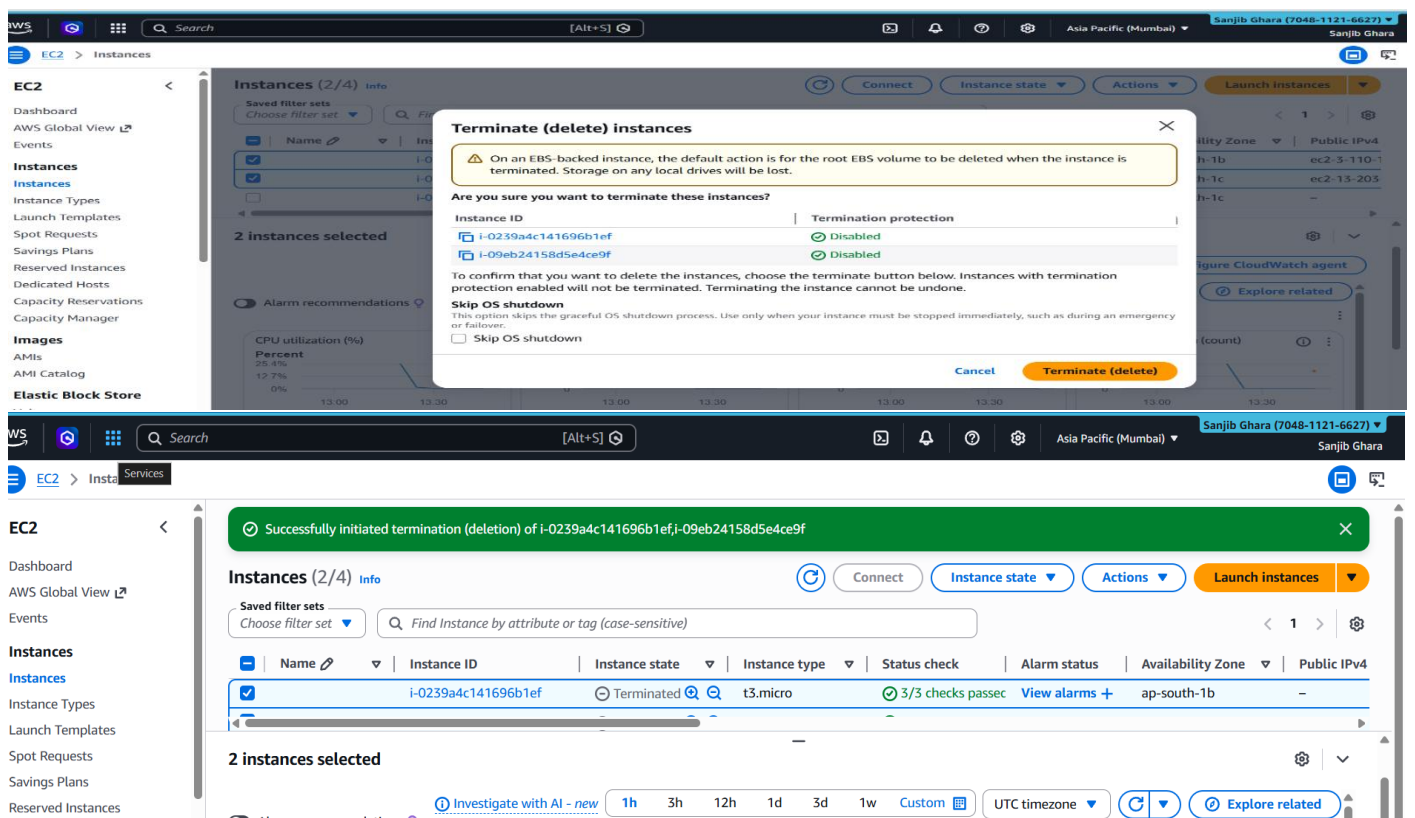


3. Delete Target Group: Go to Target Groups, select **sanas065-1**, click **Actions → Delete**, and confirm.





4. Terminate Instances: Go to the **Instances** dashboard, select any remaining running instances, click **Instance state** → **Terminate**, and confirm.



Conclusion

In this lab assignment, we successfully implemented and observed the behavior of AWS Auto Scaling and Application Load Balancing to balance the load across different EC2 instances. By configuring a Launch Template and an Auto Scaling Group, we achieved a highly available architecture capable of both self-healing and dynamic scaling.